

# Notebook

April 8, 2021

## 0.0.1 Question 1

In the context of estimating the value of houses, what does error mean for an individual homeowner? How does it affect them in terms of property taxes?

In the context of value of houses, error greatly affects how much the homeowner is expected to pay or manage in terms of housing cost. Higher value means higher property taxes. If the error is high in the value, it means that it negatively effects low income homeowners paying high taxes and values. It also positively effects high income homeowners paying lower taxes if the error is high. This creates a discrepancy in equality of homeowner to value.

## 0.0.2 Question 2

How do the CCAO's transparency initiatives aim to redistribute power between the tax lawyer industry, the CCAO, and the constituents of Cook County?

**Hint:** You may find the following [official post from CCAO](#) helpful.

Making the data open source allows for journalist and analysts to analyze the data and give potential house buyers context of how expensive a house really is through data science. In relation to tax payers, this also gives corrections to what the actual value is since there are more eyes on it, hoping to equalize the value and fairness between house owners and their respective incomes.

## 0.0.3 Question 3

Take a look at the Residential Automated Valuation Model files under the Models subgroup in the CCAO's [GitLab](#). Without directly looking at any code, do you feel that the documentation sufficiently explains how the residential valuation model works? Which part(s) of the documentation might be difficult for nontechnical audiences to understand?

From a software engineer POV, the documentation is very verbose, neat, and organized. I know everything that's going on without looking at the code. I can already notice that the underlying model is a gradient boosted decision tree. From a nontechnical POV, the files and documentation may be hard to understand, but it is generalized enough with nice visuals and graphics.

## 0.0.4 Question 4

How could the CCAO improve its implementation of transparency? What aspects of it are inaccessible and to whom? Consider the concepts of expertise and power.

CCAO can improve its transparency by providing more intuitive visualizations and graphics like it already has been doing, to make it more understandable to a wider range of audience. The technical

aspects such as `git clone`-ing the repo and running the models in R is not something anyone can do. Something more accessible (like a website) can definitely improve the interpretation and transparency to all people ranging from data scientists, city council, tax payers, and home owners.

### 0.0.5 Question 5

How does the CCAO's open data initiative support it in maintaining people's trust in its housing assessments?

It supports maintaining people's trust by showing the predicted value using a good machine learning model. It prevents house sellers from inflating or deflating the price from what the house is really truly evaluated at, giving fairness to home buyers and tax payers. Again, making the data public also ensures this trust between homeowners and the CCAO.

**Question 7a** Recall the optimal value of  $\theta$  minimizes our loss function. One way of solving for  $\theta$  is by taking the derivative of our loss function with respect to  $\theta$ , like we did in HW 6.

Write/derive the expressions for following values and write them with LaTeX in the space below.

- $R(\mathbf{x}, \mathbf{y}, \theta_1, \theta_2)$ : our loss function, the empirical risk/mean squared error
- $\frac{\partial R}{\partial \theta_1}$ : the partial derivative of  $R$  with respect to  $\theta_1$
- $\frac{\partial R}{\partial \theta_2}$ : the partial derivative of  $R$  with respect to  $\theta_2$

Recall that  $R(\mathbf{x}, \mathbf{y}, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n (\mathbf{y}_i - \hat{\mathbf{y}}_i)^2$

$$R(\mathbf{x}, \mathbf{y}, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n (y_i - (\theta_1 x_i + \sin(\theta_2 x_i)))^2$$

$$\frac{\partial R}{\partial \theta_1} = \frac{1}{n} \sum_{i=1}^n -2x_i (y_i - (\theta_1 x_i + \sin(\theta_2 x_i)))$$

$$\frac{\partial R}{\partial \theta_2} = \frac{1}{n} \sum_{i=1}^n -2x_i \cos(\theta_2 x_i) (y_i - (\theta_1 x_i + \sin(\theta_2 x_i)))$$

**Question 8d: Analyzing Learning Rates** In 1-2 sentences, describe what you notice about the path that  $\theta$  takes with a static learning rate vs. a decaying learning rate and for batch vs. stochastic gradient descent. Based on these comparisons, why do we often prefer SGD over batch GD in practice? In your answer, refer to the plots above.

Static learning rate only descends to an optimal theta by oscillating back and forth. The decaying learning rate converges slower and slower per iteration. In contrast, stochastic gradient descent oscillates and kind of traverses all around with more variance than static learning rate, until it narrows down to optimal. It still covers near the optimal value. In practice, if we are working with Big Data, where it's infeasible to compute over the whole dataset, it may be practical to use SGD in terms of performance, as we achieve near similar behavior.